

**MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY,
BHOPAL - 462003**

Name of the program		B. Tech.	Semester	IV	Year	2 nd
Name of Course:		Electronic Devices & Systems				
Course Number:		EE 223				
Core/Elective/other		Core				
Pre-requisites: Basic Knowledge of Electronics Devices and Circuits						
Course Outcomes						
1	To understand basics of OP AMPs, feedback topologies, Oscillators, Digital logic devices, combinational and sequential circuits					
2	To analyze Simple and complex Circuits of OP AMPs, feedback topologies, Oscillators, Combinational and Sequential logic circuits					
3	To design OP AMP circuit for given applications and analog computation					
4	To design combinational logic circuits and sequential circuits for a real world problems using Boolean algebra and Karnaugh Map					
Description of Contents in brief:						
1.	Differential and OP AMP basics: Differential Amplifiers their types & characteristics, OP AMP Block Diagram, OP AMP terminology, Definition of key parameters. OP AMP applications- Analysis and design of various open loop and closed loop applications. OP AMP applications in Mathematical operations & analog computers, Instrumentation amplifier, Current to Voltage and Voltage to Current converter, Schmitt trigger and Multivibrator circuits.					
2.	Feedback topologies: Introduction to Feedback amplifier's topologies. Analysis and Design of Voltage Series, Current Series, Voltage Shunt and Current Shunt topologies. Oscillators: Barkhausen Criterion, Wien Bridge Oscillator, RC Phase Shift Oscillator, Hartley, Clapp and Colpitts Oscillator..					
3.	Introduction to Digital Electronics: Logic Gates, Logic Families, Boolean algebra, Karnaugh Map upto 4 variables.					
4.	Combinational Circuits: Half/ Full Adder, Half/ Full Subtractor, Adder-Subtractor, Decoder/Encoders, Multiplexer/Demultiplexer.					
5.	Sequential Circuits: Flip flops & their Application, Registers and Counters.					
List of Text Books:						
1.	Millman & Halkies, Electronic Devices & Circuits, , McGraw Hill Education					
2.	Ramakant A. Gayakwad, Op-Amp & Linear Integrated Circuits,					
3.	William H.Gothmann, Digital Electronics, 9 edition, Pearson, July 28, 2011					
4.	Thomas Floyd, Digital Fundamentals					
5.	A. P. Malvino, Electronics Principles, McGraw-Hill Higher Education,					
List of Reference Books:						
1.	Schaum's Outline Series, Electronics Devices & Circuits,					

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2.	Morris Mano, Digital Design, Pearson Education
URLs	
	https://nptel.ac.in/courses/108/102/108102112/

1) Digital Fundamentals by Thomas Floyd

2) Digital Logic and Computer Design by Morris Mano

3) Digital Principles and Applications by Donald P Leach, Albert Paul Malvino and Goutam Saha